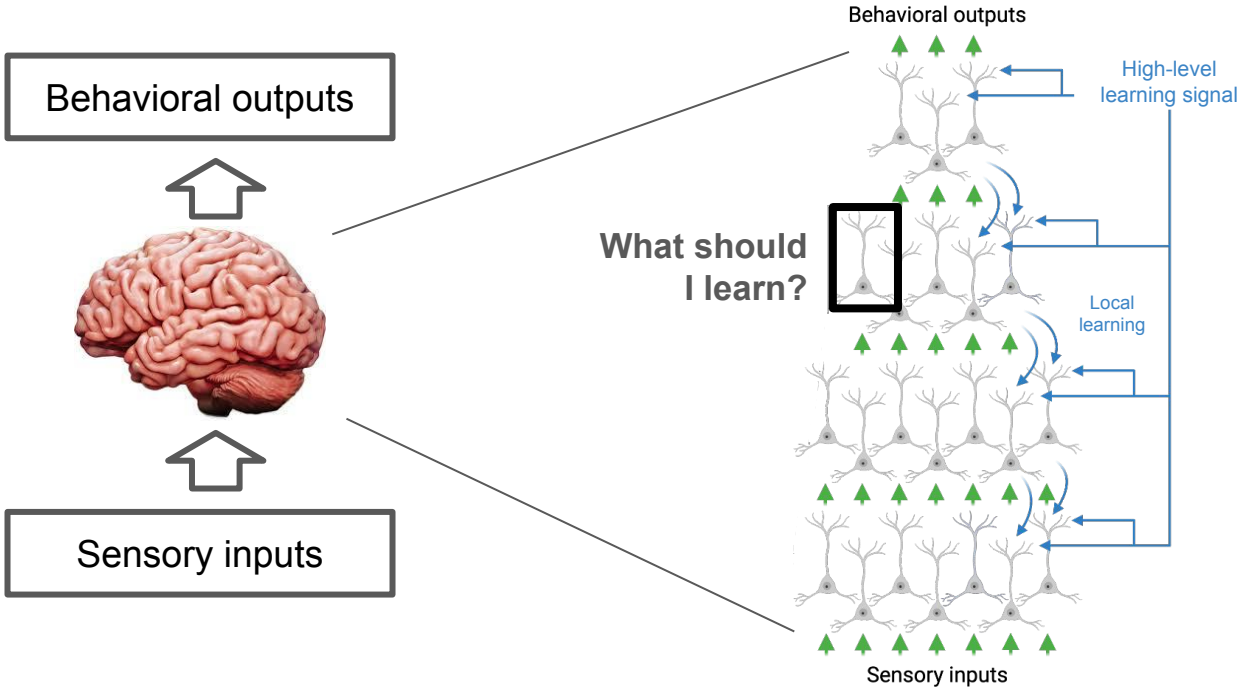
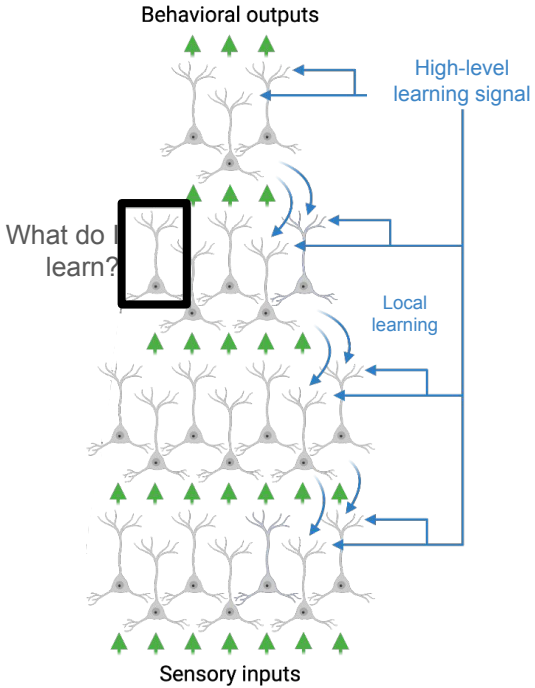
Today’s question: Credit assignment



Algorithmic credit assignment: History



- Backprop … or not
  - Rummelhart & Hinton, *Nature* 1986
  - Crick, *Nature* 1989



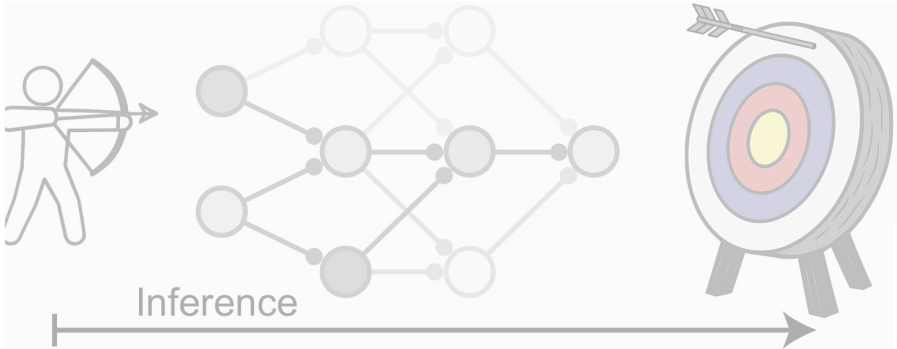
- Now many models exist for and against BP
  - Richards et al. *Nat. Neuro.* 2019
- Interest beyond academia

The Economist
August 14 2024

Quanta Magazine
February 18 2021

Scientific American
March 18 2024

Framework

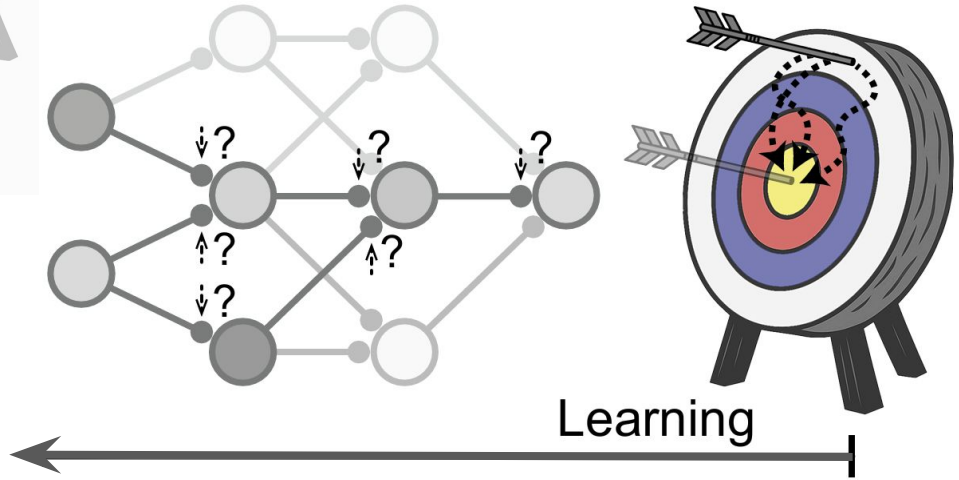


Neural activity to process an input

- Action/sensory-evoked neural responses (neuroscience)

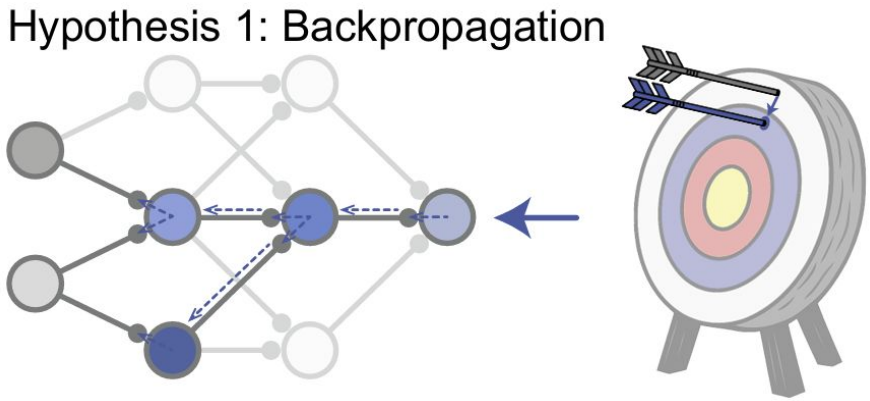
Adapt connection strength

- Through feedback (neuroscience)
- Backward pass (machine learning)



Hypothesis: Network Learning through Backprop

Backpropagation (BP): **Adapt** the current activity to reduce error



Forward pass $(h_0 = x)$

$$a_l = W_l h_{l-1} + b_l, \qquad h_l = f(a_l), \qquad \mathcal{L} = \mathcal{L}(h_L, y)$$

Error signal $\delta_l \equiv \dfrac{\partial \mathcal{L}}{\partial a_l}$

1. **Output layer**

$$\delta_L = f'(a_L) \odot \nabla_{h_L} \mathcal{L}$$

2. **Backward recursion** (chain rule)

$$\delta_l = \left(\frac{\partial a_{l+1}}{\partial a_l}\right)^\top \delta_{l+1} = f'(a_l) \odot \left(W_{l+1}^\top \delta_{l+1}\right)$$

3. **Gradients** (a_l is linear in W_l, b_l)

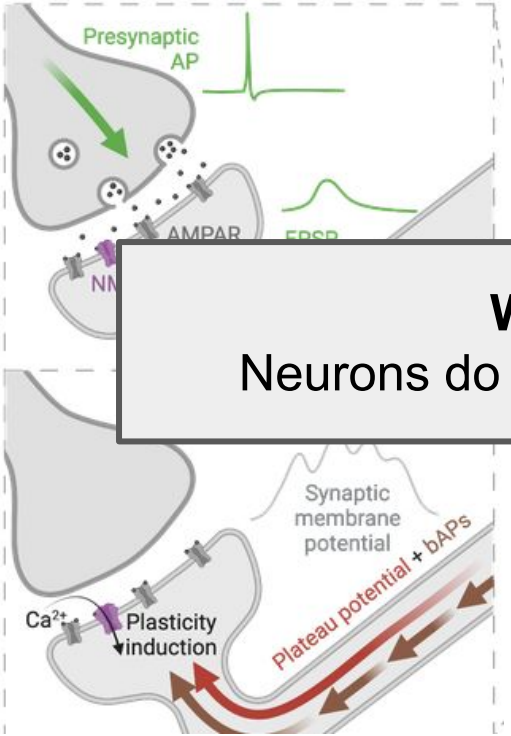
$$\frac{\partial \mathcal{L}}{\partial W_l} = \delta_l \, h_{l-1}^\top, \qquad \frac{\partial \mathcal{L}}{\partial b_l} = \delta_l$$

4. **Update**

$$W_l \leftarrow W_l - \eta \underbrace{\delta_l}_{\text{post: error}} \underbrace{h_{l-1}^\top}_{\text{pre: activity}}$$

Model circuits: Sacramento et al. (+Senn) 2017, Whittington & Bogacz 2017, Scellier & Bengio 2017, Payeur et al. 2021

Why is backpropagation considered implausible



Synaptic transmission

- Presynaptic neuron releases Glutamate
- Postsanyptic neuron opens calcium channels
- he soma

Weight transport problem:
Neurons do not have forward-backward symmetry!

- NMDA open + depolarization lets Ca2+ in the spine
- Ca2+ gets into the postsynaptic spine
- The calcium triggers kinases/phosphatases

But do you really need weight symmetry?

Architecture & Dataset	Backpropagation (BP)	Direct Feedback (DFA)	Feedback Alignment (FA)
Fully Connected (FC) *Dataset: MNIST*	**1.45%**	1.45%	1.47%
Fully Connected (FC) *Dataset: CIFAR-10*	**45.1%**	45.4%	46.1%
Convolutional Network (CNN) *Dataset: CIFAR-10*	**23.1%**	27.2%	27.3%

Source: towards data science, A. Jimenez

What if we project the errors backwards with random weights?

*and initially zero forward weights

It works…

Lillicrap, Timothy P., et al. "Random synaptic feedback weights support error backpropagation for deep learning." *Nature communications* 7.1 (2016):
Nøkland, Arild. "Direct feedback alignment provides learning in deep neural networks." *Advances in neural information processing systems* 29 (2016).

Why would that ever work?

Setting: linear net $y = W_2 W_1 x$, error $e = y - y^*$, $W(0) = 0$, gradient flow; $\langle \cdot \rangle$ = average over data.

1. What we need. The FA hidden signal $\delta_{\text{FA}} = B\,e$ is a descent direction (per sample) iff

$$\langle \delta_{\text{FA}},\, \delta_{\text{BP}} \rangle = (Be)^\top (W_2^\top e) = e^\top W_2 B\, e \;>\; 0$$

2. Random B imprints itself on the forward weights.

$$M(t) \equiv -\int_0^t \langle e\, x^\top \rangle\, ds \qquad \text{(error–input correlation)}$$

$$\dot{W}_1 = -B\langle e\, x^\top \rangle = B\dot{M} \;\Rightarrow\; W_1 = BM$$

$$\dot{W}_2 = -\langle e\, x^\top \rangle W_1^\top =$$

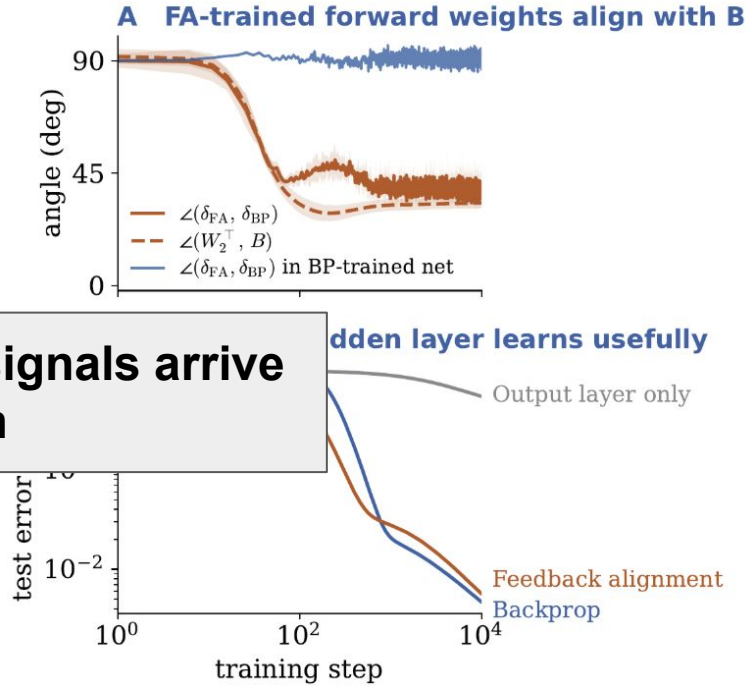
3. Hence alignment

$$e^\top W_2 B\, e \approx c\, e^\top N e$$

using $N + N^\top = \int (\dot{M}M^\top + M\dot{M}^\top)\, ds = MM^\top$.

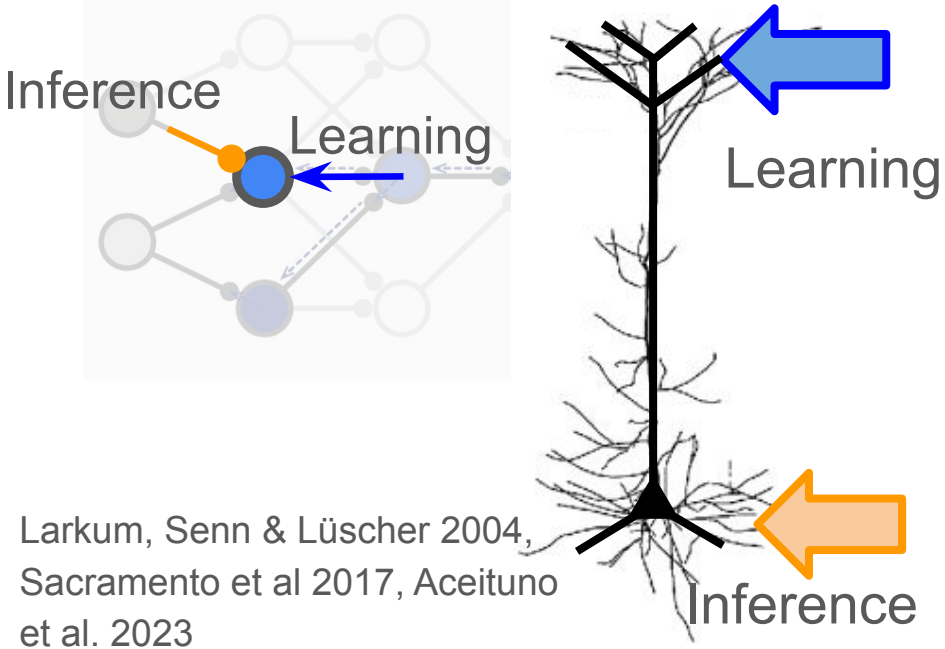
How would the learning signals arrive to the neuron

Summary: First learn to align the forward weights, then use BP



30–20–10 tanh net, linear teacher, SGD from small init; mean ± s.d., 5 seeds.

Test synaptic plasticity rules



Larkum, Senn & Lüscher 2004, Sacramento et al 2017, Aceituno et al. 2023

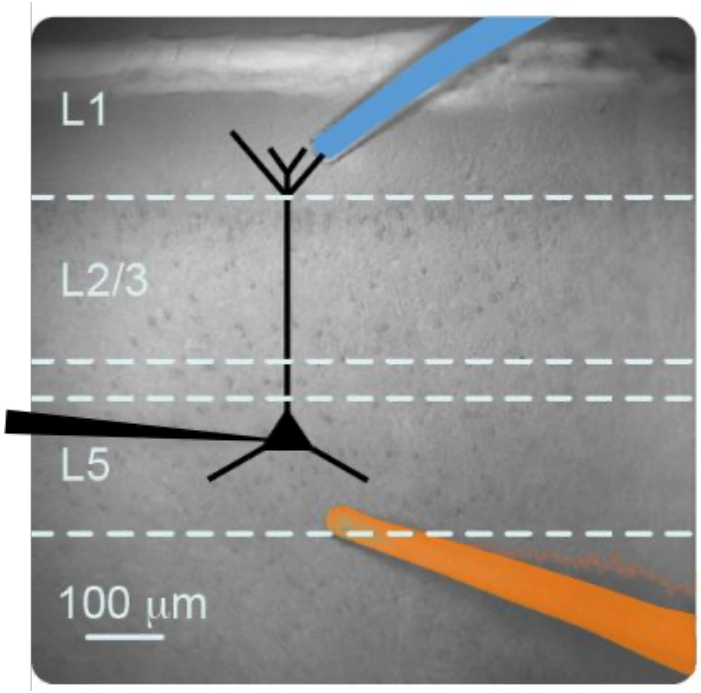
Apical inputs…

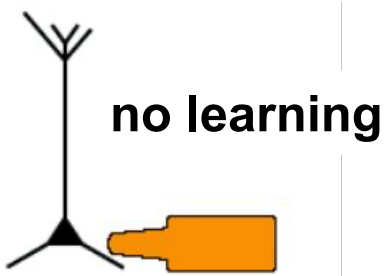
- Induce plateaus
- Provide inputs from distal areas

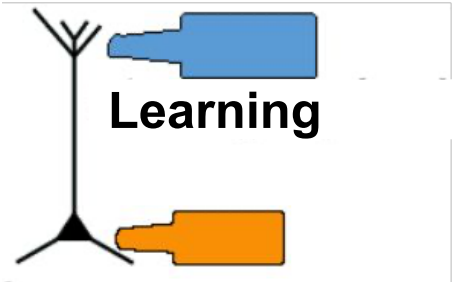
$$\Delta w_{pre \rightarrow post} \propto a_{pre} \times \delta_{post}$$

Experiment by
R. Loidl (INI)

Electrophysiology set-up







Experiment:

Neurons:

- Layer 5 neurons
- Prefrontal area
- 12 neurons, 6 mice

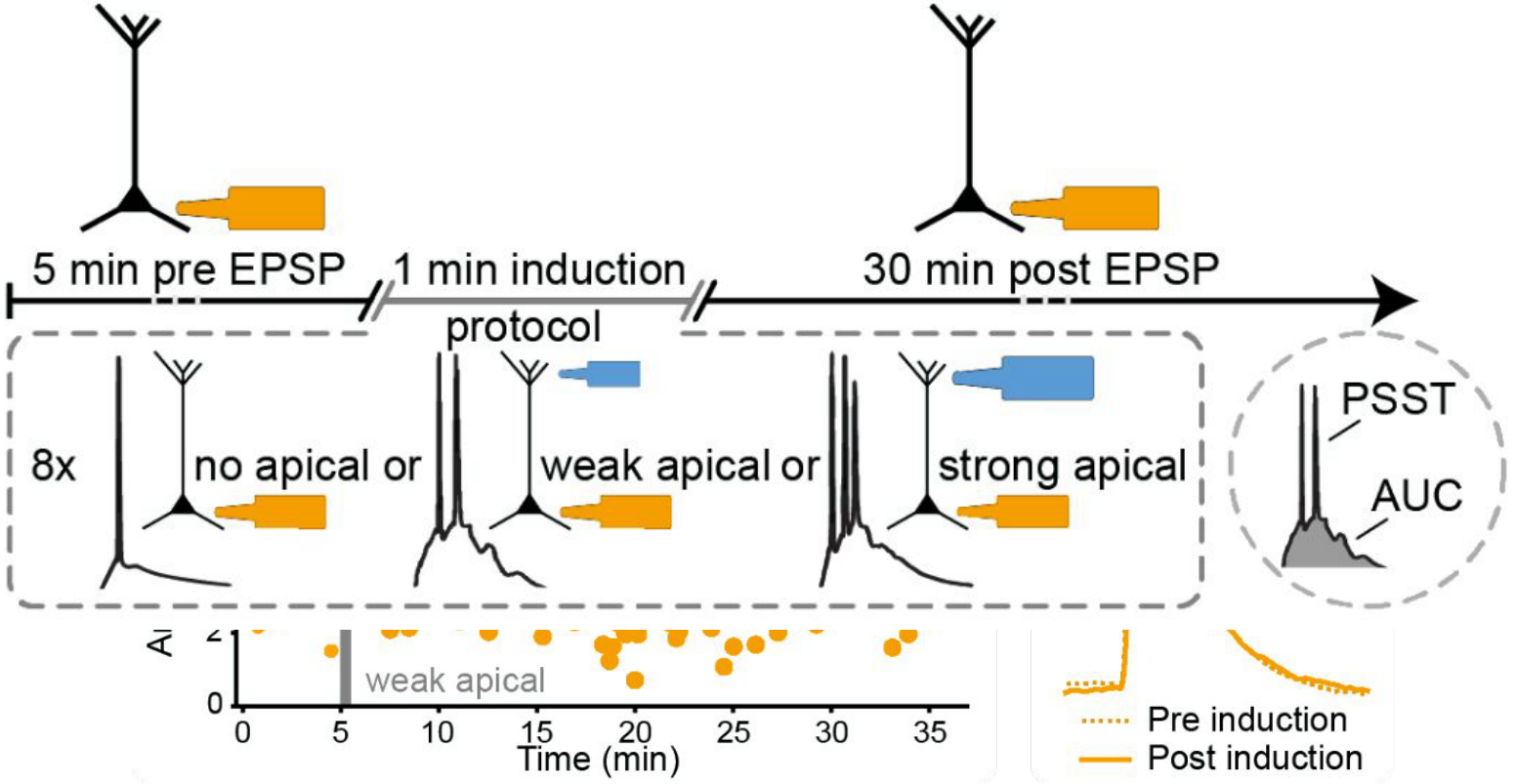
Measurement

- Patch clamp soma
- Extracellular electrodes

Protocol

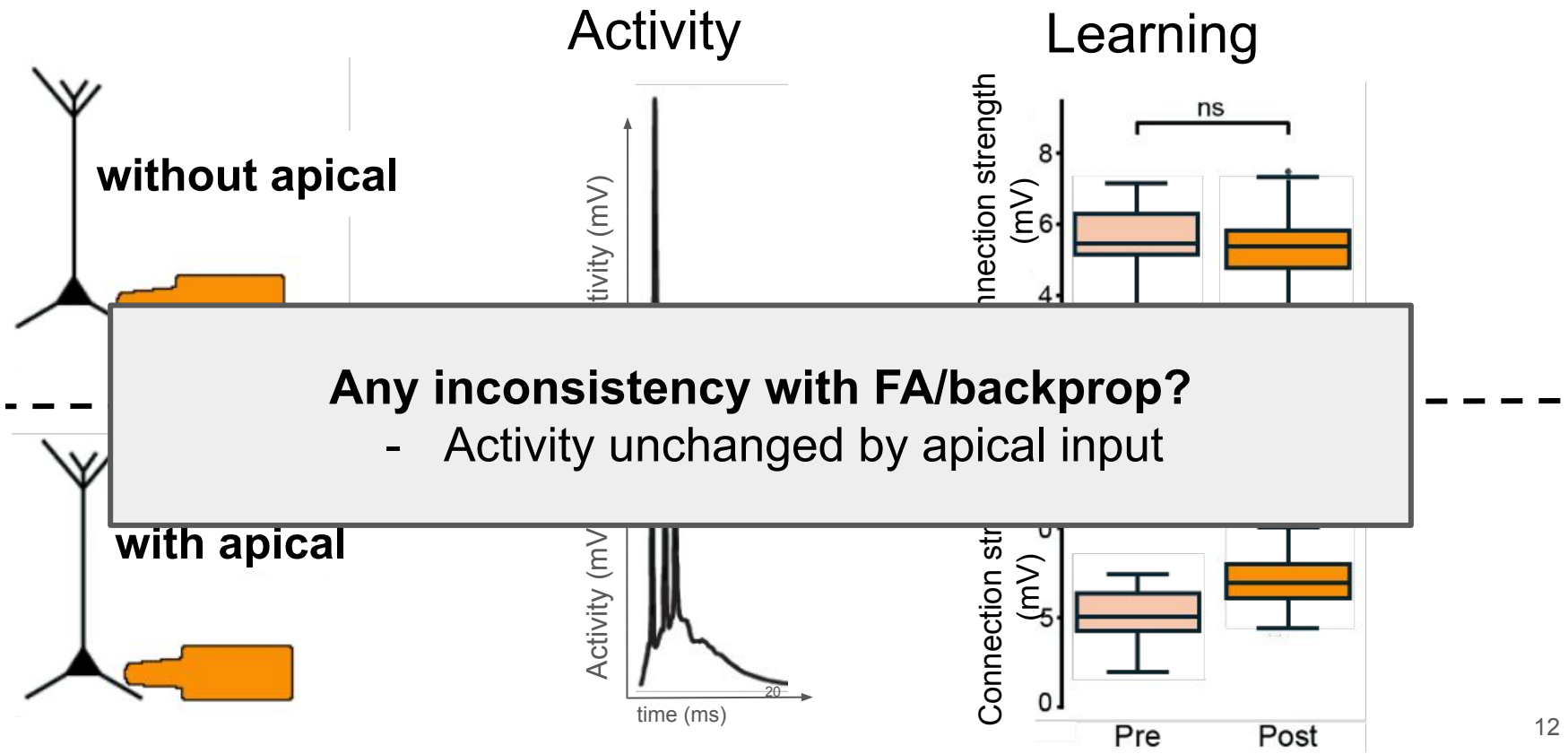
- Measure EPSP
- 8 repetitions
- Measure again

Experiment & Data

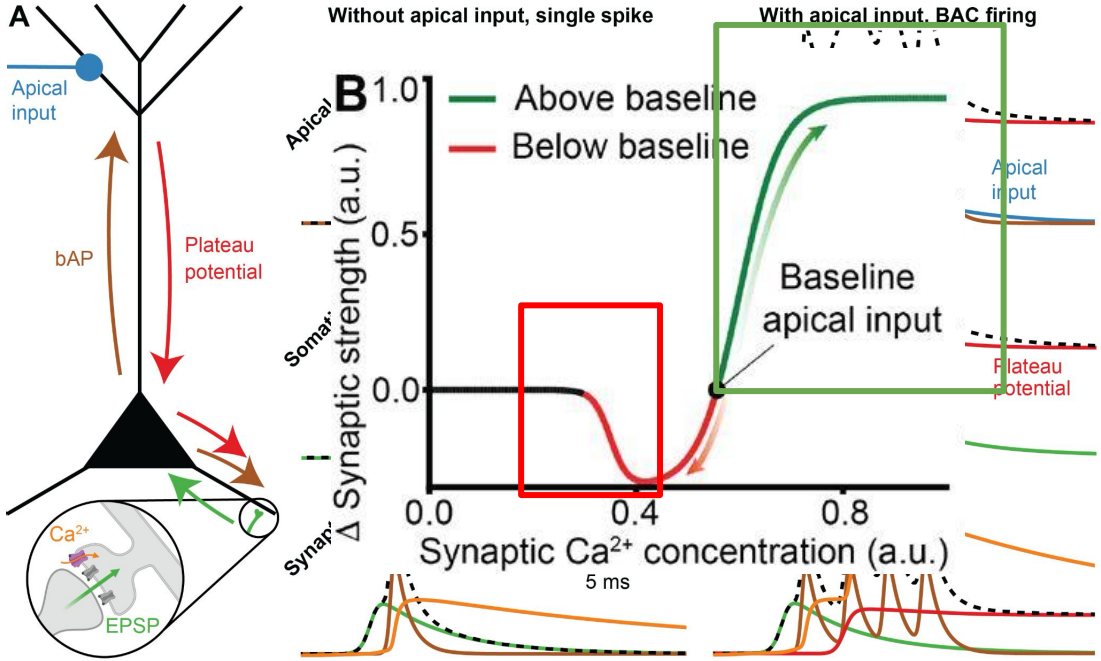


Experiment by
R. Loidl (INI)

Electrophysiology: apical inputs induce plasticity



Neuron model: from calcium to learning and bursts



Calcium in the synapse

Backpropagating AP

- Somatic AP
- Dendritic plasticity
  - Ingelbert et al. PNAS 2020
  - Graupner & Brunel PNAS2012

Plateau potentials

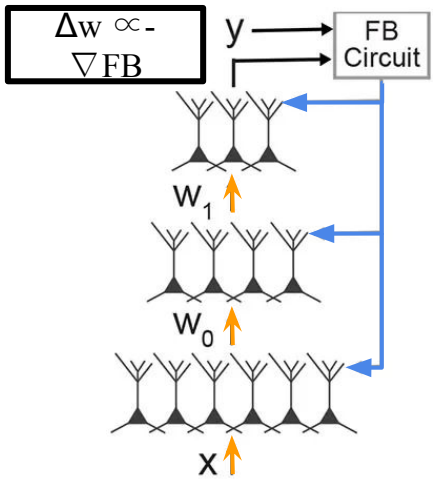
- How to get calcium in?
- Requires somatic AP and **apical input**
- Long lasting depolarization

Bursts

- From plateau potentials
- Plus other inputs

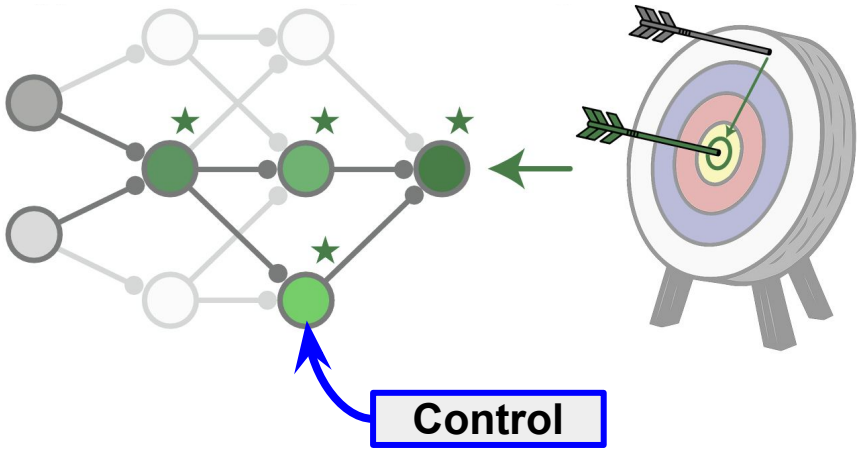
Clopath, Claudia, and Wulfram Gerstner. "Voltage and spike timing interact in STDP–a unified model." *Frontiers in synaptic neuroscience*

Another alternative: prospective/target methods



Rao & Ballard, Neur. Comp. 1999
Meulemans et al. NeurIPS 2021
Song et al. Nat. Neu. 2024

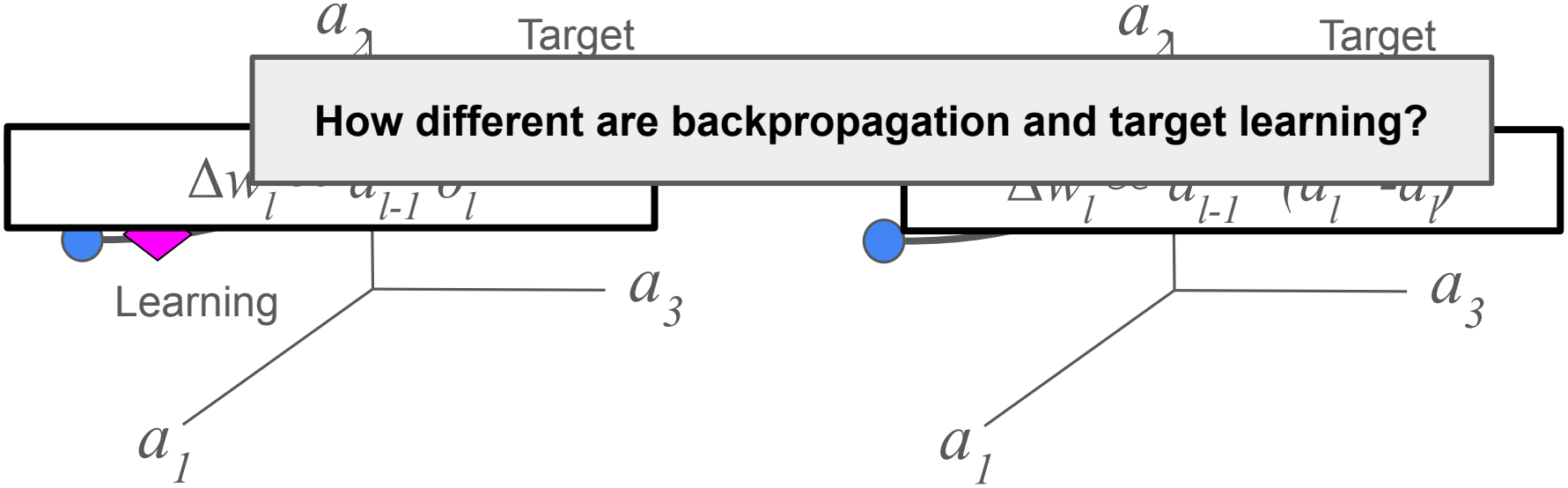
Target Learning (TL): **Enforce** the desired activity, learn to decrease the forcing



Hypotheses: Network Learning Algorithms

Backpropagation & co: work on the inference activity and reduce error

Target Learning (TL): **Enforce** the desired activity, learn to decrease the forcing



How different are backpropagation and target learning?

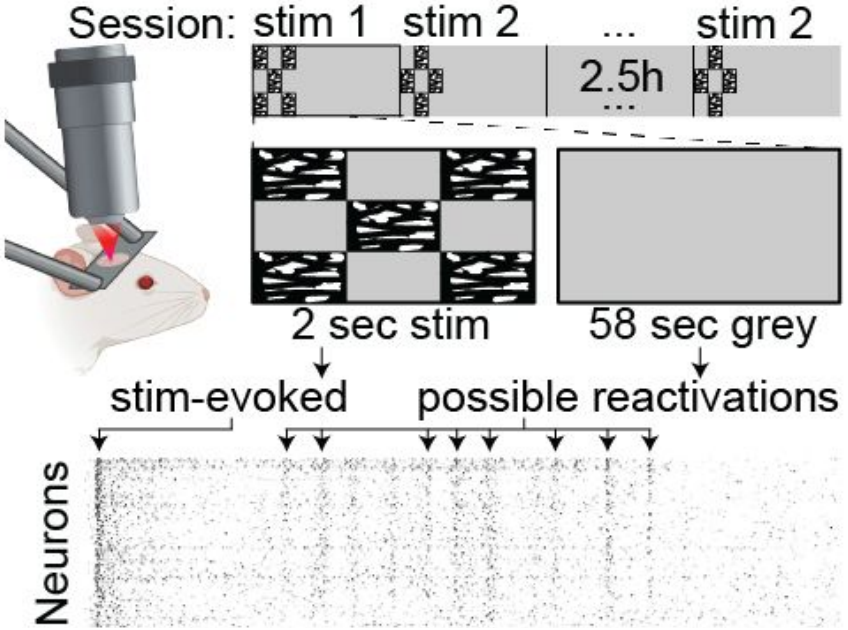
Learning in vivo - available data

Data desiderata

- Recordings during learning
- Separation of learning and inference

Cortical reactivations: Recurrence of stimulus-evoked activity without stimulus

- Needed for learning
- Happen without stimulus

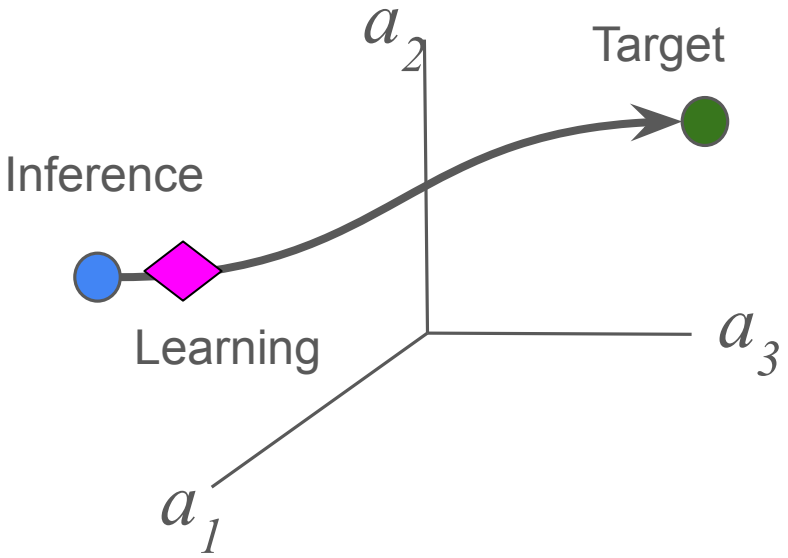


Nguyen et al 2024: Calcium imaging of mouse lateral visual cortex during stimulus presentation and reactivations

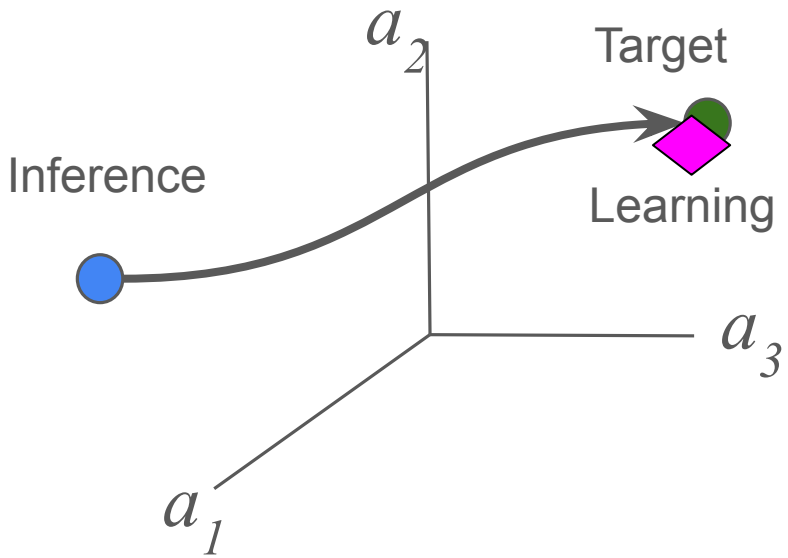
Source: Anderman Lab (Harvard)

Hypotheses: Network Learning Algorithms

Backpropagation & co: work on the inference activity and reduce error



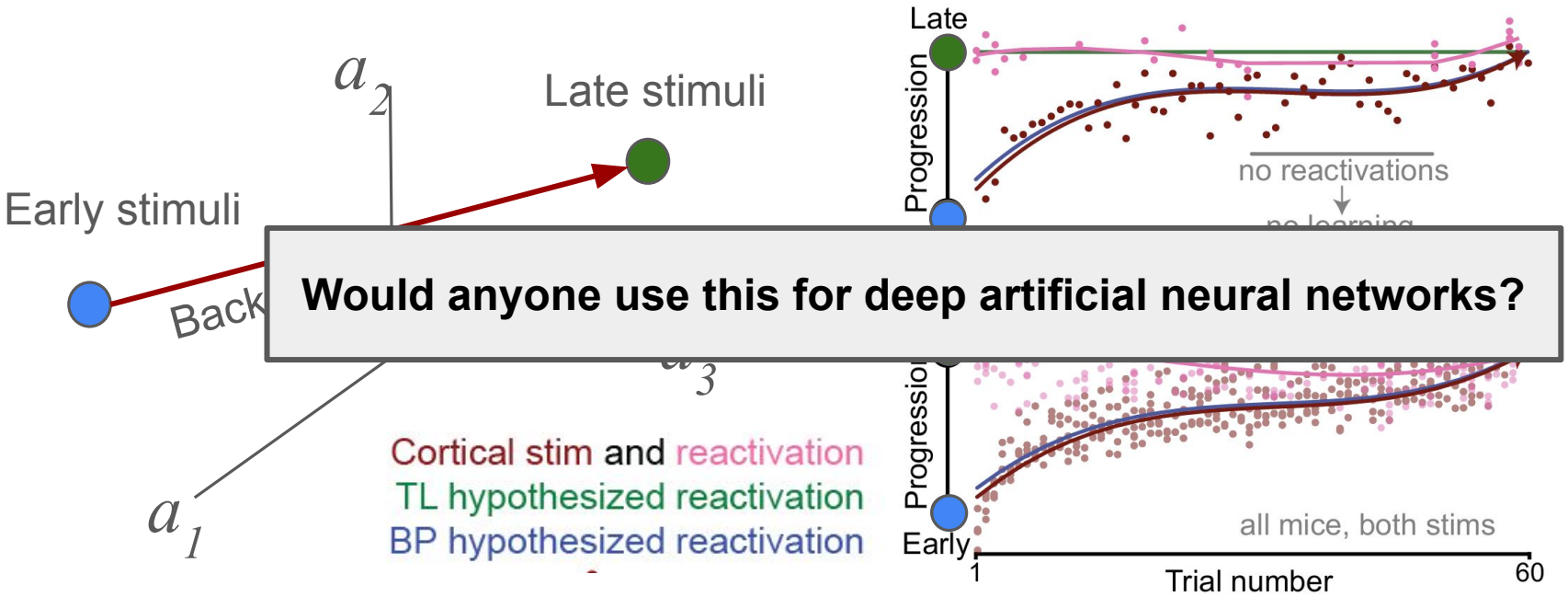
Target Learning (TL): **Enforce** the desired activity and adapt synapses to decrease the forcing



Test algorithmic hypotheses



Sander de Haan (INI)



P. V. Aceituno, S. de Haan, L. Beumer, R. Loidl, B. Grewe “Challenging backpropagation: evidence for target learning in the neocortex”, under review

How do we compute the target?

Forward: $h_l = F_l(h_{l-1}) = f(W_l h_{l-1} + b_l)$, $h_0 = x$. Feedback: learned $G_l \approx F_l^{-1}$.

1. Output target (small step downhill)

$$\hat{h}_L = h_L - \hat{\eta}\, \nabla_{h_L} \mathcal{L}$$

2. Propagate targets (difference TP)

$$\hat{h}_l = G_{l+1}(\hat{h}_{l+1}) + \left[\, h_l - G_{l+1}(h_{l+1}) \,\right]$$

bracket cancels inverse errors: $\hat{h}_{l+1} = h_{l+1} \Rightarrow \hat{h}_l = h_l$

3. Local update (regress onto target)

$$W_l \leftarrow W_l - \eta\, \underbrace{f'(a_l) \odot (h_l - \hat{h}_l)}_{\text{replaces } \delta_l}\, h_{l-1}^{\top}$$

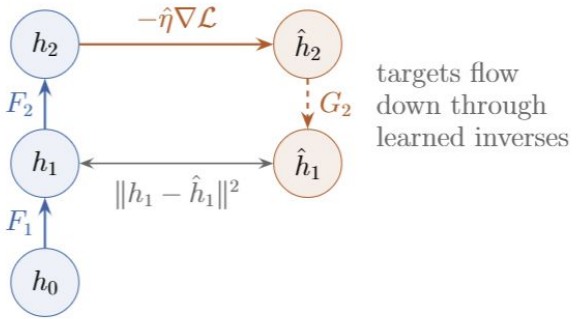
4. Train the inverse (local, with noise ϵ)

$$\min_{G_l} \left\| G_l\big(F_l(h_{l-1} + \epsilon)\big) - (h_{l-1} + \epsilon) \right\|^2$$

What it computes (linearised)

$$\hat{h}_l - h_l \approx -\hat{\eta}\, \underbrace{J_{G_{l+1}} \cdots J_{G_L}}_{\approx J_{F_{l+1:L}}^{-1}} \nabla_{h_L} \mathcal{L} \qquad \text{BP: } J_F^{\top} \text{ instead of } J_F^{-1}$$

$\Rightarrow$ Gauss–Newton step on activations, not gradient descent on weights.



targets flow down through learned inverses

Lineage

2014 **TP** (Bengio): targets via autoencoder inverses
2015 **Difference TP** (Lee et al.): correction term, works on MLPs
2018 Bartunov et al.: falls short of BP on large conv nets
2020 Meulemans et al.: TP $\approx$ Gauss–Newton; better inverse loss
2022 Ernoult et al.: targets matching BP, scales further

Pros local updates, no weight transport; target view of learning; GN-like conditioning; no f' needed to send targets.

Cons needs good inverses (fails for bottlenecks); extra feedback training; not gradient descent on W; lags BP at scale.

A multi-scale problem

